

Brandon Chhom

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EDUCATION

University of California, Santa Cruz, B.S. Computer Science

Expected Graduation June 2028

GPA: 3.55/4.0, Dean's Honors List

Relevant Courses: Analysis of Algorithms, Computer Systems & C Programming, Data Structures & Algorithms, Machine Learning, Mobile Applications, Software Design Project

EXPERIENCE

Software Engineer Intern | IDX Exchange (Real Estate)

Remote | Jun 2026 - Sep 2026

- Reduced property-search query latency ~1,700x (14.7s → 8ms) by analyzing MySQL EXPLAIN plans and adding targeted indexes, verifying each one's real impact before keeping it
- Implemented sorting and filtering for a 53,000-row property API using parameterized queries and a whitelisted-column approach, eliminating SQL-injection risk across 8 sort combinations
- Built a full-featured React property search app — image gallery/lightbox, Google Maps integration, localStorage-backed favorites — covered by 200+ automated Jest/RTL tests

Software Engineer Intern | Selene Odds (Sports Betting Analytics)

Remote | Jun 2026 - Sep 2026

- Shipped a full-stack feature for browsing live Kalshi prediction-market odds, organizing 1,000+ markets into a category/sport/event hierarchy inferred directly from the data
- Integrated Kalshi as a new live odds source and surfaced prediction markets through site navigation and search, enabling side-by-side comparison with existing betting lines
- Designed an extensible multi-venue pricing model enabling additional sportsbooks and prediction markets without UI schema changes

Python Tutor | UCSC Baskin Engineering

Santa Cruz, CA | Jan 2025 - Jun 2025

- Tutored 30+ one-on-one and small-group sessions on Python fundamentals, data structures, OOP, and debugging
- Simplified complex programming concepts through targeted explanations and debugging walkthroughs, reducing common student misconceptions

PROJECTS

[GudEats](#) - Social content platform | MongoDB, Express.js, React.js, Node.js

Jun 2026 - Jul 2026

- Engineered a secure REST authentication system in Node.js/Express using JWT, httpOnly cookies, and bcrypt password hashing with protected-route middleware
- Integrated Cloudinary and Clarifai's computer-vision API to auto-validate that uploaded photos contain food, designing the check to fail open on third-party outages so a vision-service failure never blocks uploads.
- Implemented a calendar-week-scoped social feed with UTC date-boundary calculations and paginated queries, validated against live database inserts across week-boundary edge cases (Sunday/Monday rollover)

[ThinkBack](#) - AI-Powered Active Learning Platform | React, Node.js, Express.js

Jan 2026

- Built an AI learning platform that generates real-time questions and study guides from video lectures using React and Node.js
- Built a scalable Node.js/Express backend with WebSocket support to generate real-time questions via the Opennote API
- Extracted video context by integrating OpenAI Whisper for transcription and OCR for slide-text extraction

[UCVibe](#) - Campus Review Platform | React.js, Node.js, Firebase

Apr 2025 - Jun 2025

- Built a React-based multi-campus review platform with campus-specific facility data, dynamic facility pages, review submission flows, and reusable UI components across dining halls, cafes, markets, lecture halls, and recreation centers

SKILLS

Languages: Python, TypeScript, JavaScript, C, SQL, HTML/CSS

Frameworks & Libraries: React.js, Next.js, Express.js, Pandas

Technologies & Tools: Node.js, Git, GitHub, Docker, Firebase, MongoDB, MySQL